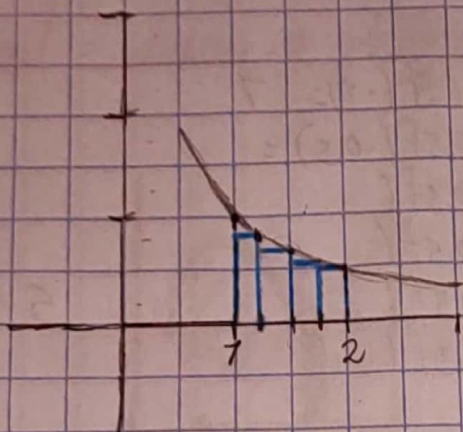


Tarea sección 5.1

3) a) $f(x) = \frac{1}{x}$

de $x = 1$ a $x = 2$

$$\Delta x = \frac{b - a}{n} = \frac{2 - 1}{4} = \frac{1}{4}$$



$$A_1 = \left(\frac{1}{4}\right) f(1.75) = \left(\frac{1}{4}\right) \left(\frac{4}{5}\right) = \frac{1}{5}$$

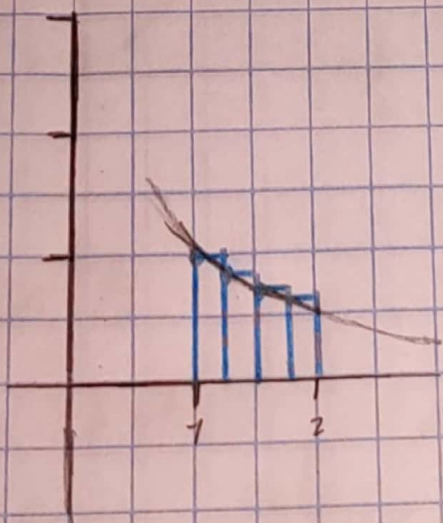
$$A_2 = \left(\frac{1}{4}\right) f(1.5) = \left(\frac{1}{4}\right) \left(\frac{2}{3}\right) = \frac{1}{6}$$

$$A_3 = \left(\frac{1}{4}\right) f(1.25) = \left(\frac{1}{4}\right) \left(\frac{4}{5}\right) = \frac{1}{5}$$

$$A_4 = \left(\frac{1}{4}\right) f(1) = \left(\frac{1}{4}\right) (1) = \frac{1}{4}$$

$$A_T = A_1 + A_2 + A_3 + A_4 = \frac{1}{5} + \frac{1}{6} + \frac{1}{5} + \frac{1}{4} = \frac{533}{840} \quad \text{subestimación}$$

b)



$$A_1 = \left(\frac{1}{4}\right) f(1.75) = \left(\frac{1}{4}\right) \left(\frac{4}{5}\right) = \frac{1}{5}$$

$$A_2 = \left(\frac{1}{4}\right) f(1.5) = \left(\frac{1}{4}\right) \left(\frac{2}{3}\right) = \frac{1}{6}$$

$$A_3 = \left(\frac{1}{4}\right) f(1.25) = \left(\frac{1}{4}\right) \left(\frac{4}{5}\right) = \frac{1}{5}$$

$$A_4 = \left(\frac{1}{4}\right) f(1) = \left(\frac{1}{4}\right) (1) = \frac{1}{4}$$

$$A_T = \frac{1}{5} + \frac{1}{6} + \frac{1}{5} + \frac{1}{4} = \frac{379}{420} \quad \text{sobrestimación}$$

13)



$$\text{Subestimación} = (0 \times 0.5) + (1.4 \times 0.5) + (3.3 \times 0.5) + (4.5 \times 0.5) + (5.5 \times 0.5) \\ (5.9 \times 0.5) = \underline{10.55}$$

$$\text{Subestimación} = (1.4 \times 0.5) + (3.3 \times 0.5) + (4.5 \times 0.5) + (5.5 \times 0.5) \\ (5.9 \times 0.5) + (6.2 \times 0.5) = \underline{13.8}$$